

Integrative Multi-omics Dissection of Epigenetic and 3D Genome Regulatory Logic in Immune-Hot and Immune-Cold Hepatocellular Carcinoma



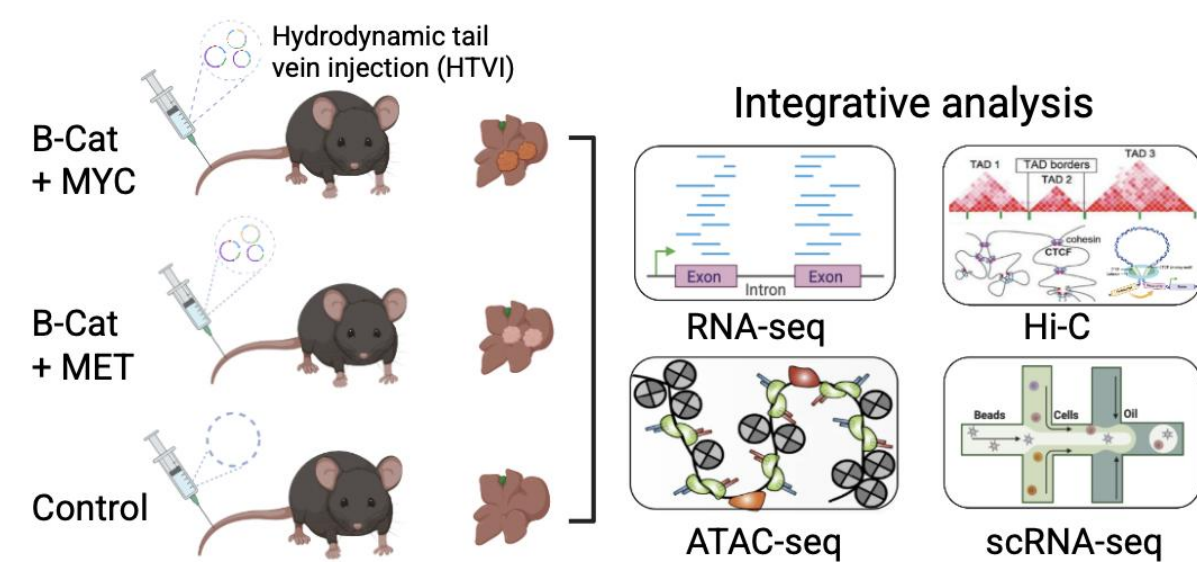
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Background

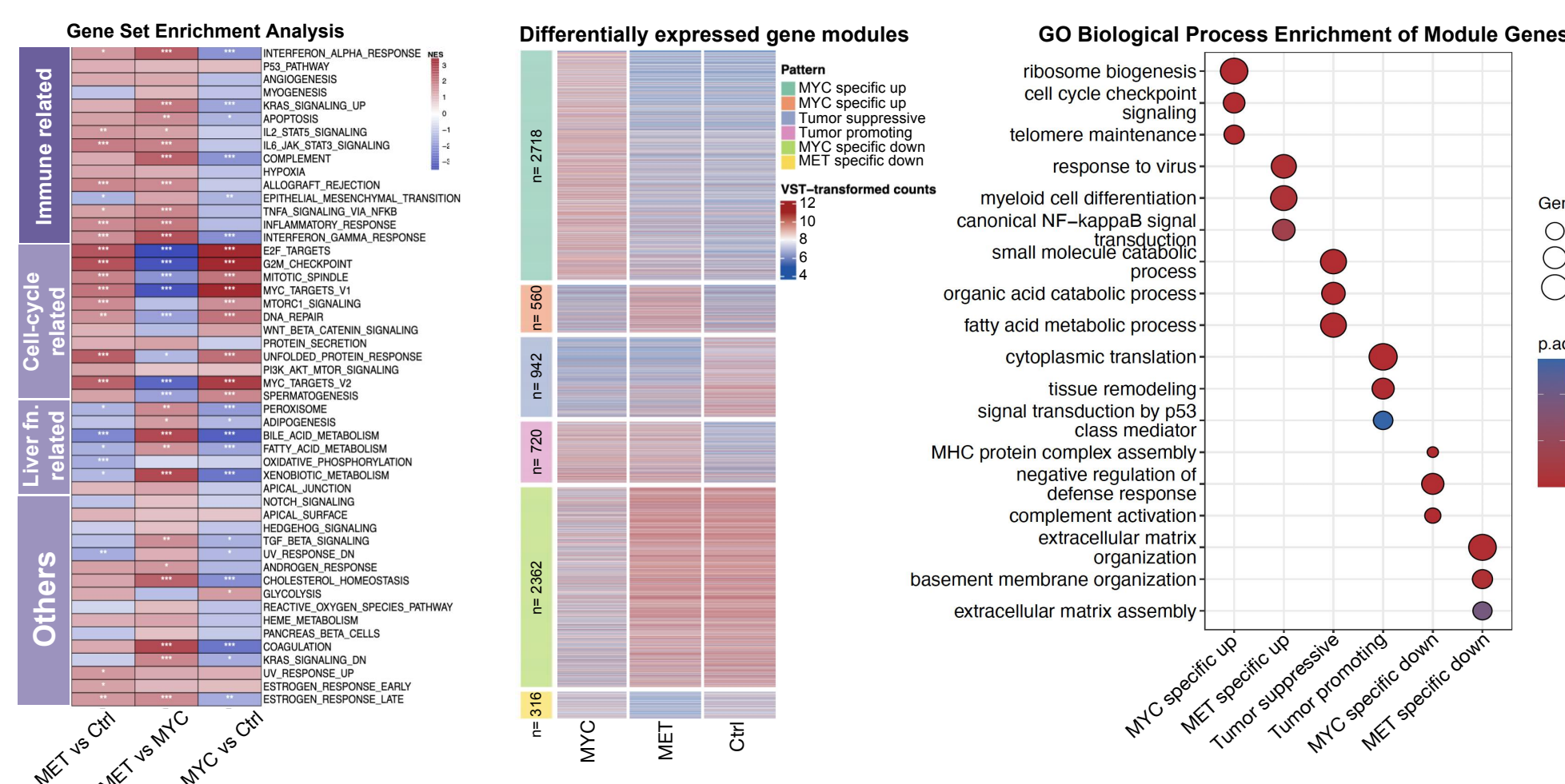
Hepatocellular carcinoma (HCC) is a highly lethal malignancy characterized by rising global incidence and limited therapeutic efficacy. Clinically, HCC tumors often present as either immune-infiltrated ("hot") or immune-depleted ("cold"), which strongly impacts treatment response. In this study, we found that our spontaneous mouse models (MET + β -catenin and MYC + β -catenin) faithfully recapitulate these clinically relevant "hot" and "cold" phenotypes, respectively. Emerging evidence indicates that the three-dimensional (3D) genome architecture provides a critical spatial framework for epigenetic regulation and gene expression. However, the 3D genome landscapes of hot versus cold tumors, as well as the molecular mechanisms underlying their differences, remain poorly understood.

Overview of the study design



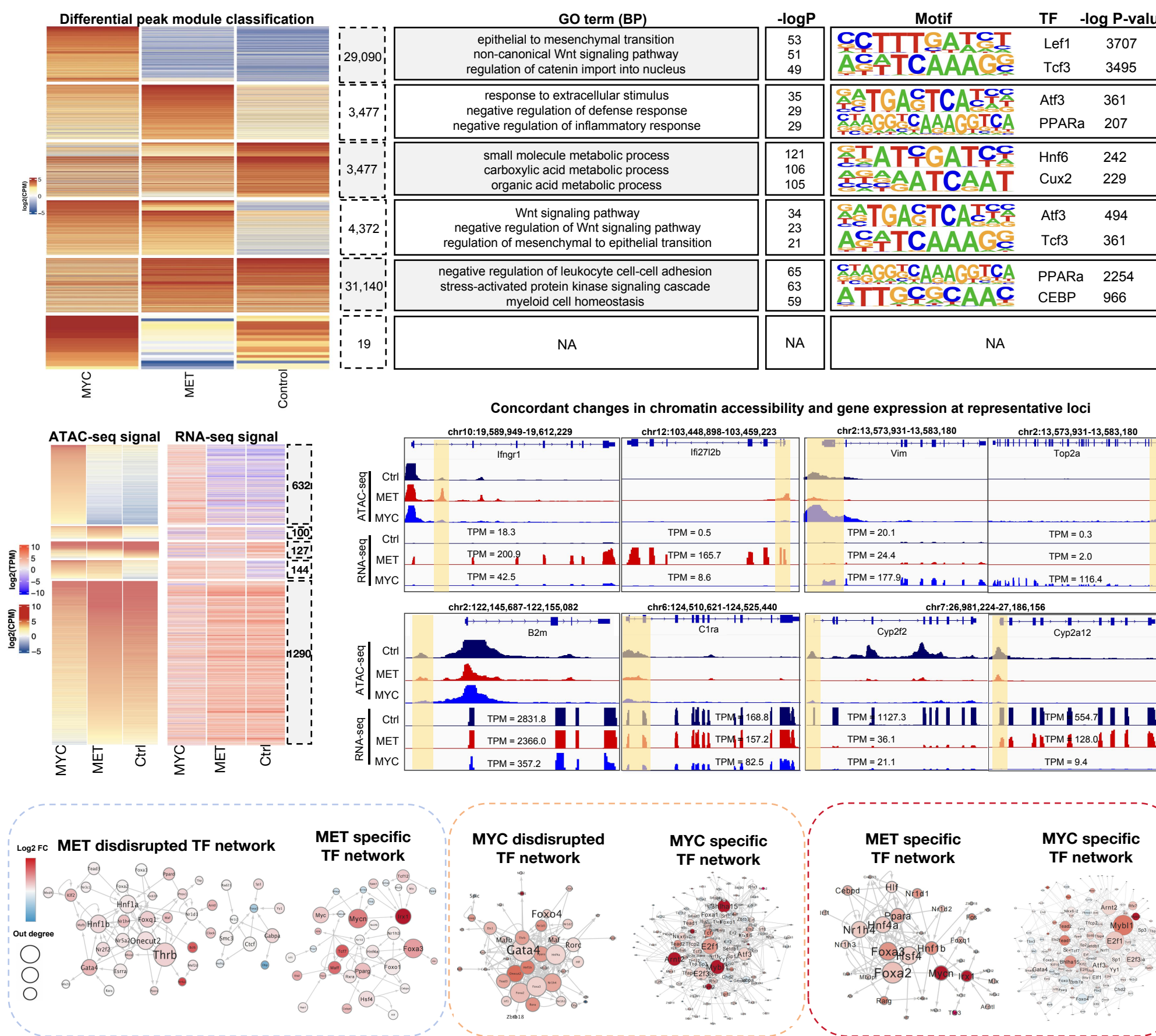
We established MYC- and MET-driven HCC mouse models and performed an integrative multi-omics analysis combining Hi-C, ATAC-seq, and RNA-seq to profile 3D genome structure, chromatin accessibility, and transcriptional outputs.

Distinct transcriptional states in MET- and MYC-induced hepatocellular carcinoma

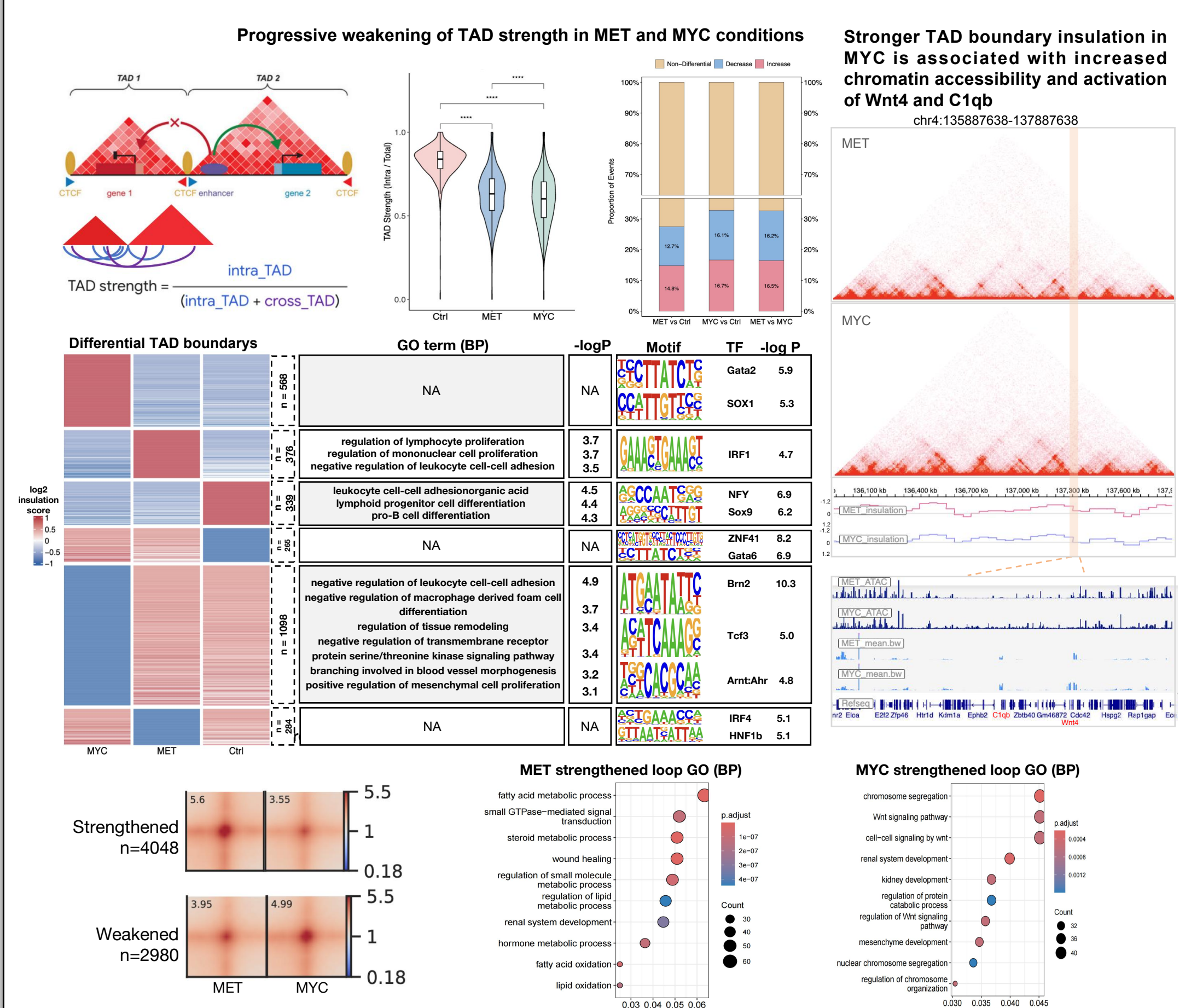


GSEA reveals that MET- and MYC-driven tumors exhibit distinct immune states, with MET associated with immune activation and MYC with immune suppression. These differences are driven by divergent gene expression programs regulating pathways such as innate immunity and cell cycle progression.

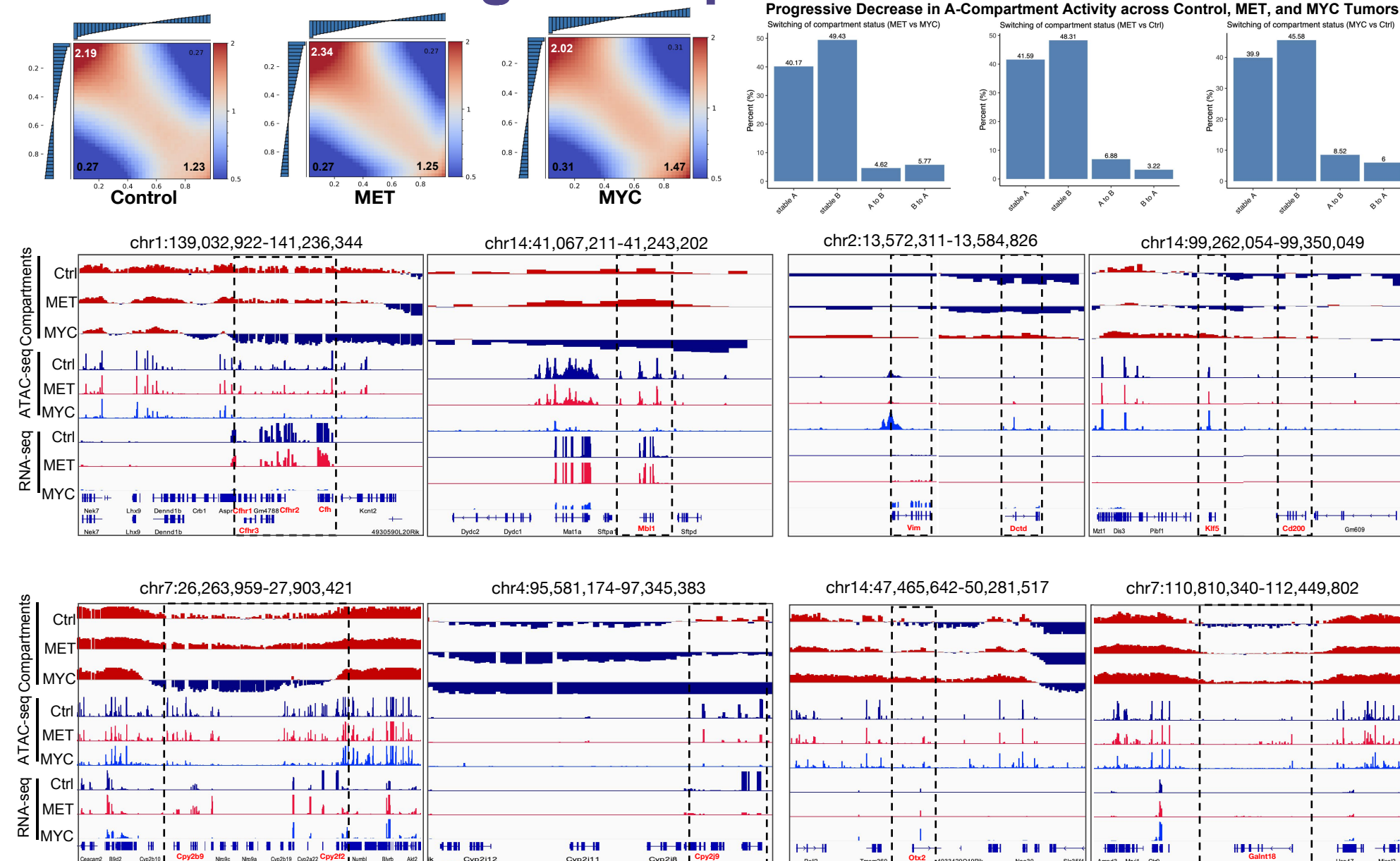
Key transcription factors coordinate chromatin accessibility to drive transcriptional programs



3D chromatin remodeling underlies oncogene-specific regulatory programs



Global A/B compartment switching regulates gene expression



Conclusion

Our integrative multi-omics analysis reveals that oncogene-specific rewiring of the 3D genome architecture dictates the divergent epigenetic landscapes and transcriptional programs underlying HCC heterogeneity.

Acknowledgement

NIH P30 CA054174, NIH S10 OD030311, NIH R50 CA265339-01A1, CPRIT First-Time, Tenure-Track Faculty Award (RR240016), UT Rising STARS (UT175673), MCC-American Cancer Society Early Career pilot award.